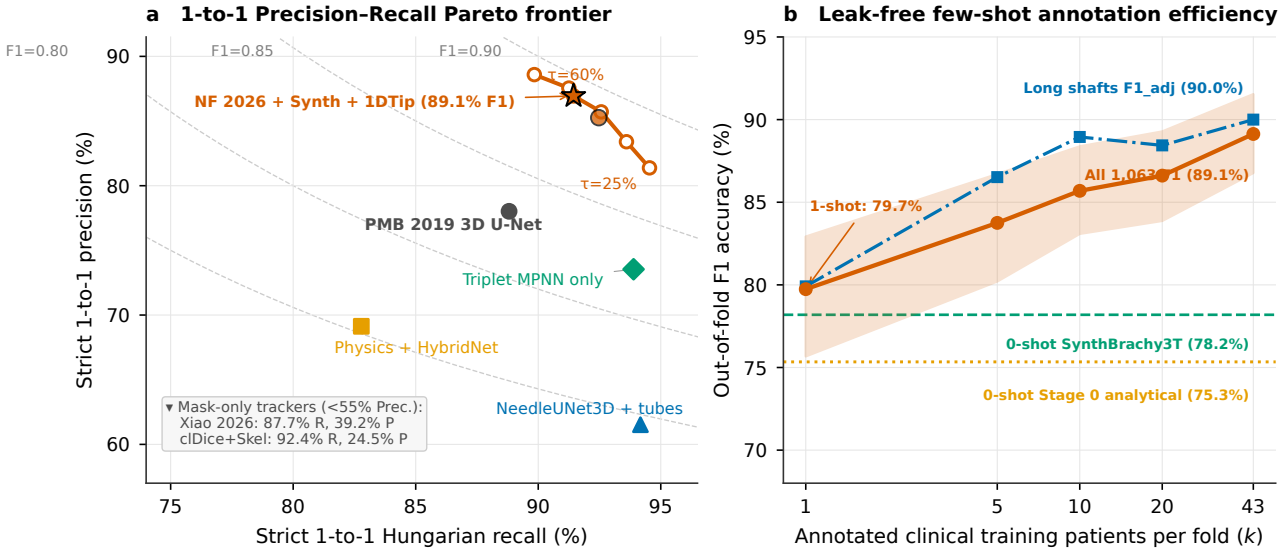
a 1-to-1 Precision–Recall Pareto frontier
F1=0.80
F1=0.85
F1=0.90
τ=60%
NF 2026 + Synth + 1DTip (89.1% F1)
τ=25%
PMB 2019 3D U-Net
Triplet MPNN only
Physics + HybridNet
NeedleUNet3D + tubes
▾ Mask-only trackers (<55% Prec.):
Xiao 2026: 87.7% R, 39.2% P
clDice+Skel: 92.4% R, 24.5% P
Strict 1-to-1 precision (%)
Strict 1-to-1 Hungarian recall (%)
90
80
70
60
75
80
85
90
95
b Leak-free few-shot annotation efficiency
Long shafts F1_adj (90.0%)
All 1,063 F1 (89.1%)
1-shot: 79.7%
0-shot SynthBrachy3T (78.2%)
0-shot Stage 0 analytical (75.3%)
Out-of-fold F1 accuracy (%)
Annotated clinical training patients per fold (k)
95
90
85
80
75
70
1
5
10
20
43